GlucoTwin System architecture

Fusing the health record with wearable data in a personal digital twin to warn of glucose spikes and hypoglycaemia two hours ahead.

Team YETTO · VIT Vellore
Happiest Health Digital Twin Challenge, Phase 1

01 Data sources

Synthetic Indian cohort

Health record
Demographics, labs, diagnoses, medicines, genetics

Wearables
CGM every 5 min; heart rate, HRV, steps, sleep

Diary app
Meals, medicine doses, walks

02 Data fusion

One 15-minute timeline

Time alignment
All streams on a single clock, no future information

Feature set
75 features with CGM, 69 without

Outcome labels
Spike above 180 or hypo below 70 mg/dL within 2 h

03 Digital twin

Personal physiology model

Calibration
Fits each patient's insulin action and secretion (week 1)

Physiology model
Bergman minimal model with Indian meals and 5 drug classes

Forecast
Glucose 2 to 4 hours ahead from any moment

04 Intelligence

Predict, explain, simulate

Spike predictor
XGBoost with a GRU fusion network

Hypo detector
Alerts when the twin's forecast falls below 75 mg/dL

Explainer
TreeSHAP: the reasons behind each alert

What-if simulator
Compares two care plans on the twin

05 Clinician dashboard

Decision support

Patient and twin
Record, time in range, personal parameters

Early alerts
Spike and hypo risk 2 hours ahead, with reasons

Plans and evidence
What-if comparison, cohort, model validation

DATA
100% synthetic: 200 patients, 14 days each (DPDP Act 2023, HIPAA)

VALIDATION
Twin calibrated on week 1, tested on week 2 with CGM hidden; models scored on unseen patients

STACK
Python, XGBoost, PyTorch, Streamlit